

AI-Generated or AI-Modified? User Reactions to Labeling AI Use in Social Media Posts

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Abstract

Although social media platforms are beginning to implement policies for labeling AI usage in posts, we do not know how this labeling affects user perceptions and which types of labeling source and language would be most effective. What does AI-generated content mean to users? What are users' perceptions of the content, content creator, and platform that has AI-labeled content? Do the effects differ depending on whether the label is attributed to the user or the platform? A focus group study (N=14) revealed that users appreciate how AI helps to create better content. However, their perceptions of AI-labeled content are shaped by their mental models of how social media algorithms work. Some participants viewed AI-labeled content as more trendy, while others saw it as direct advertisements. Some believed the content to be automatically fake, but their reactions varied depending on the type of content or account. Regarding labeling source, users preferred self-labeling over platform labeling. We discuss theoretical and practical implications for the design of social media interfaces for disclosing AI usage.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

AI Labeling, UGC, User-Generated Content

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1 Introduction

Much of the online world, especially social media, runs on user-generated content (UGC). There are 34 million TikTok videos being uploaded daily, about 272 videos every second [13]. Also, over 100 million photos and videos are uploaded to Instagram daily [14]. With the arrival and rapid diffusion of generative AI, the notion of UGC has undergone a fundamental change. Many of the things posted by users are increasingly being created by AI entirely or with AI's help in some way (such as editing, retouching, etc.), thus raising doubts about the true source of content. Open AI reported that in 2022, users generated more than 2 million images per day with DALL-E 2, an image-generation model [16], implying that a large portion of social media content could be AI-generated without users' notice. In order to address this question in users' minds, popular platforms that rely on UGC are either asking users to disclose whether they used AI in their post or implementing tools to automatically flag suspected AI-created content, so that the consumers of the content are forewarned about the potential involvement of AI in the posts they encounter. However, it is not known whether platform users even pay attention to such labels attributing AI as the source, and if they do, whether they care. Does the involvement of AI somehow diminish the authenticity or integrity of the content posted by another user? Or, does it enhance the perceived quality and credibility of the content because the user has taken the effort to incorporate emergent technological tools? Are the effects any different if the person posting the content discloses AI involvement compared to the platform flagging a post as potentially AI-derived? This study set out to answer these questions by conducting a focus group with 14 participants who use social media platforms such as Facebook, Instagram or TikTok.

2 Related Works

In this section, we survey current AI labeling practices and users' perceptions of AI-labeled content.

2.1 AI Labeling Practices

People often find it difficult to distinguish whether the content is generated by AI or humans. One suggested solution to this issue is

to clearly and openly label AI-generated content. This strategy has already garnered attention from lawmakers both in the U.S. and internationally [21]. At a White House summit in July 2023, major tech companies committed to creating strong technical systems to inform users when content is AI-generated. These labeling efforts are already appearing across platforms[2]. Several social media platforms have started making efforts to indicate whether content is generated or modified by AI. Meta was the first to introduce the 'Made with AI' label for content created or modified by AI [3]. Later, they updated this policy, replacing it with the 'AI Info' label to avoid misleading users into thinking content that was slightly edited by AI was fully generated by AI. Instagram also introduced the 'AI Info' label, and additionally, it specifies whether the label is provided by the platform or the poster [7]. TikTok uses "AI-generated" flag to declare that the content is generated by AI. The notion of AI labeling is not entirely new, as social media platforms have long utilized various types of content labels, such as fact-check labels, warning labels for sensational content, and disclaimer labels for advertising.

2.2 User Perception on AI Usage on the Content

Previous research on AI usage in content has yielded diverse findings. Studies have primarily focused on the effect of AI usage on users' credibility perceptions of the content. AI-sourced content is perceived as less credible [25] and more artificial than human [28]. News labeled as AI-generated is rated as less trustworthy than news without such a label [24]. This is related to algorithmic aversion, which refers to a negative bias in both behavior and attitudes toward content generated by algorithms, as opposed to human-generated content, even when the algorithm's output is objectively superior. However, some research shows the opposite effect of AI usage on content credibility. For example, articles created by a computer were perceived as more credible and scored higher in terms of journalistic expertise [6]. Another line of research has shown how AI usage affects users' content evaluations, such as perceived objectivity or beauty depending on the context. For example, one study found that individuals consider AI-generated paintings to be less beautiful than those created by humans [20]. However, other studies have found no effects of an algorithmic creator on perceptions of content objectivity [10][28]. Recent research shows that labeling content in this way may not achieve the intended goal of helping users assess its credibility. Instead, it could negatively impact the creator or give users a false sense of confidence in the accuracy of the information [19]. In addition to these contradictory findings, the potential impact of AI labeling on user-generated content (UGC) within the social media ecosystem remains under-explored. While previous studies have focused on users' perceptions of individual content, whether generated by AI or humans, the effect of AI labeling in UGC needs to be understood in the context of the broader social media environment. Following the introduction of misinformation labels on Facebook in 2016, unexpected results emerged. Users, being familiar with such labels, found that in the absence of a tag, the lack of a misinformation label can create an 'implied truth' effect, making the content appear more credible to users [17]. In a study about the effect of AI-mediated communication on online self-presentation, Jakesch found that users' perception

of AI-generated profiles of Airbnb hosts varies depending on the circumstance. When people are presented with only AI-generated profiles, they trust them just as much as they would trust human-written profiles [8]. However, when people are shown a mix of AI- and human-written profiles, they tend to mistrust hosts whose profiles they believe were generated by AI. Using the Hyperpersonal Model of CMC, the authors argued that receivers often place excessive emphasis on signals from the sender's self-presentation because of the limited cue in text-based CMC [8] [26]. We propose that adding AI labeling to user-generated content (UGC) on social media could influence users' perceptions in a similar way, as the presence of a flag would make users more aware that some content is AI-generated. Therefore, we are investigating users' general perceptions of AI-generated content and their perceptions of AI-labeled content on social media separately. It is important to note that adding an AI label may not have the same impact as users simply knowing that the content is AI-generated.

2.3 Source of AI-Labeled Contents

Considering that individuals evaluate a message based on the credibility of its source, how users perceive the actual source of AI-labeled content is important. The identity of the source can be difficult to determine under multiple layers of attribution. If users see AI-labeled content posted by an influencer on their Instagram account, is the source the account owner, the AI, or Instagram itself? According to Kang et al. (2011), sources can be divided into proximal sources (news aggregator portal, e.g., Google News) and distal sources (news organization, e.g., New York Times). Readers' perceptions of a news story's credibility are shaped not only by the distal source (news organization) to which the story is attributed, but also by the proximal source (news portal) through which the story is presented. When individuals are not motivated to carefully process a message, they are likely to attribute online information to the source cue that is most proximate [9]. In our study, we extend this to AI as a potential source and explore users' source attribution when they encounter content that is labeled as generated or aided by AI.

2.4 Different Types of AI Labeling

Given that the informal format or language of AI labeling is important, we also want to investigate how this labeling language affects users' perceptions. While there is no standard language, style, or format for labeling, in the field of explainable AI (XAI), preferred practices often focus on providing transparency while remaining easy to understand [5]. Similarly, in advertising endorsements, the U.S. Federal Trade Commission recommends using "clear and conspicuous" disclosure language to ensure that consumers can easily identify sponsored content as advertising [4]. User perceptions vary in response to different labeling approaches. Studies have found that greater transparency in labeling, particularly when providing details about how AI-powered systems generate outputs, is associated with higher levels of trust in the system [11][12]. However, tension exists in AI-assisted content creation. Despite the benefits of transparency labeling for AI functionality, users still tend to favor human involvement in tasks such as content classification [15] and human authorship in content creation, including news writing

Table 1: Demographics of participants

No	Age	Gender	Occupation
P1	27	Female	Professional (marketer)
P2	37	Male	Professional (marketer)
P3	45	Male	Professional (data scientist)
P4	40	Male	Professional (HR)
P5	30	Male	Professional (Retail)
P6	35	Female	Professional (Creative)
P7	24	Female	Ph.D. student
P8	28	Female	Post Doc
P9	26	Female	Ph.D. student
P10	19	Male	Undergraduate student
P11	32	Female	Ph.D. student
P12	21	Female	Undergraduate student
P13	25	Female	Ph.D. student
P14	21	Female	Undergraduate student

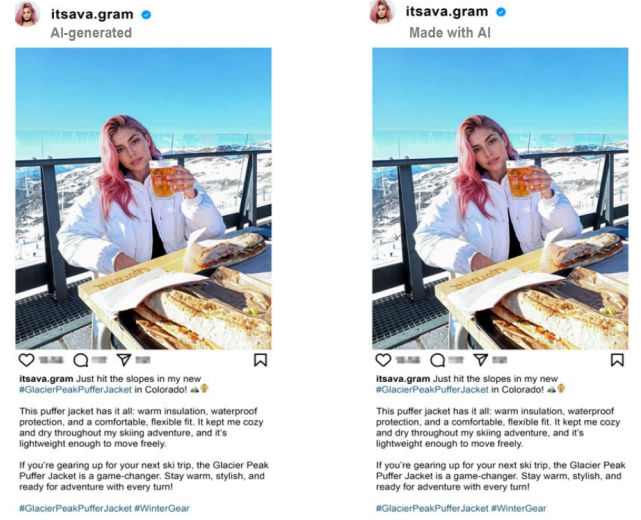
[27] and influencer marketing [29]. Preferences for labeling that indicates human involvement or identifies humans as the source may stem from a desire to restore a sense of human agency, as opposed to relying on machine agency [23]. Therefore, we assume that users will perceive AI labeling differently based on the linguistic features used to emphasize agency assignment. For example, the label “AI-generated” appears to ascribe the action entirely to AI, whereas “AI-modified” suggests some degree of human agency in the process. With advancements in technology enabling automatic flagging of such labels, we also assume that the entity deciding to apply the AI label, whether it is generated automatically by the platform’s system or disclosed by the content creator, may influence users’ sense of agency, thereby shaping distinct perceptions.

Thus, we arrived at four research questions:

- What does AI-generated content mean to users?
- How does AI-labeling affect users’ perception toward the content, content creator and platform?
- Which linguistic approach do users prefer as AI labeling among “AI-generated”, “AI-modified”, “AI Info”, “AI-created” and “Made with AI”? Why?
- “AI-generated” by humans (Disclosure) vs. by platform (Detected) –Which one do users prefer and why?

3 Focus Group Method

AI labeling in social media content is a relatively new phenomenon. Given that some participants might not be very knowledgeable about it, we wanted to explore their perception of AI-labeled content both when asked and when exposed to an actual sample. To avoid restricting their responses to pre-determined variables, we decided to conduct a focus group. We conducted five focus groups with a total of 14 participants in January 2025, either via Zoom or in person. Participants were recruited through the researcher’s personal network, social media, and a university email list. Eligible participants were individuals who had used social media platforms such as Facebook, Instagram, or Twitter. Since AI labeling on social media is not yet widespread, participants who had ever encountered content they believed was generated by AI, or that had been flagged

**Figure 1: AI Labeling examples****Figure 2: Different types of AI labeling languages**

as AI-generated or AI-modified, were also eligible to join the study. The participant’s pool was diverse in terms of age, gender, and occupation (Table 1). We developed a focus group protocol based on our research questions and the literature on AI labeling. We asked participants about their perceptions of AI-generated content. After showing AI-labeling examples (Figure 1), we asked their perception of AI-labeled content, as well as their preferred language for labeling (Figure 2). Two authors independently coded the data and then reconvened to discuss the similarities and discrepancies in their findings.

4 Results

4.1 Users’ perception of AI-generated content

Our first research question explored the meaning of “AI-generated content” for users. Many participants talked about the creation procedure for AI-generated content and emphasized human involvement in the resulting content.

4.1.1 Humans take the lead with initiative, while AI assists in enhancing human capabilities. Many users answered that AI generation means AI is helping humans in some way when creating something, but the extent of the help varies. P1 commented, “If I see that something is AI generated, it’s like someone had a very specific idea for a video or an image in mind, and then they wrote some prompt, and the image is essentially generated by AI.” P3 also mentioned “It means that it is generated from AI’s help. For example, if I want to write in a certain direction, it will reflect my intention.” Another participant thinks that it goes beyond human prompts and creates

unexpected outcomes. Example quote from P6: *“Even though I don’t have something concrete of what I want to make, or what I want to produce, if I just say it and give some more detail into it, then it gives me a whole new different outcome, so AI generation means something helping my imagination.”*

4.1.2 Is it assumed to be Fake? Some participants mentioned that ‘AI-generated’ means ‘fake’ or ‘not accurate’. Some considered that AI generated content is automatically fake as commented by P2 *“This would not be an accurate thing, or something that person would generate based on his imagination.”* One participant said that depending on the format of the content, it could be absolute fake or not. An example quote from P13: *“If they’re based in, like my LinkedIn photos, AI generated, it’s absolutely fine. But for written content or for a caption, I’d be more skeptical of it. But I don’t automatically assume that it’s completely false.”* Another participant P12 similarly noted: *“You know, it’s not necessarily fake because it is based on something like the AI machine has to reference things that we’ve made and things that are human made.”*

4.2 The effect of AI labeling on users’ perceptions of content, content creators, and platforms

To narrow the scope to AI labeling, we asked participants whether AI labeling affects their perception of the content, the content creator, and the platform. Not all participants provided a specific answer to all three components, but some participants made a distinction in their responses. The reason behind the mixed answers may be that some individuals think more deeply (e.g., about the content creator) when using social media, while others focus solely on enjoying the content itself. Their answers reflected users’ perception of AI labeling in the specific context of the social media ecosystem, such as how SNS algorithm works and how the platform adds advertisements.

4.2.1 Admitting AI-labeled content as it is, but different reaction: just enjoying, admitting trend or being cautious. Four out of 14 participants treated AI-labeled content as unofficial or not real, but their reactions to the content varied: some enjoyed it as casual content, others saw it as trendy or commercial, while some tried to be more cautious. Two participants admitted that it is not official content and tried to simply enjoy the content, as commented by P1 *“I try to get back a step and then try to look at it not as an actual thing or reality, and just to enjoy it as a fun thing.”* P2 said, *“I don’t mind the AI content itself. I think a lot of it is very creative, very artistic.”* Other participants considered AI labeled content as trendy or commercial content, as noted by P3 *“If the content is flagged as AI-generated, I may think that it is popular or trending. Thus, I feel that it might be more commercial as well.”* Another participant, P4, said that s/he tries to be careful not to be deceived by commenting: *“I should pay attention to this when I see it.”*

4.2.2 Users’ perception depends on the content or account. Several participants mentioned that their perception of AI labeled content is not fixed but dependent on the content or the account. P2 mentioned that if the content is a vacation spot where real experience matters, he won’t like AI-generated content. However, if it is showing an AI fashion show with AI model and real outfit, he does not mind

AI-generated content. P12 mentioned, *“It also depends on what the content is and what the account is. If an influencer uploaded AI generated content, I will think that the influencer is very lazy and I might want to disconnect him/her.”* P5 also commented *“There will be some accounts we expect more on the integrity. For example, we would not expect photographers or photo editors to use the AI labeling very often because we want their integrity and their actual work. But, on the other hand, like for guys who run social media accounts for just comedy, we would expect fun, funny things or funny content from that account so it wouldn’t really affect.”*

4.2.3 Negative feelings toward the labeled content but for different reasons. Some participants showed negative feelings toward AI labeled content but for different reasons. Because participants do not know why and how AI is used for content creation, they had some doubts. One reason was unknown author intention, as noted by P9: *“I would just scroll down and like ignore that because I kind of do not want to reply to that because I don’t know what the author really want to express with it if the content is generated by AI.”* Another reason is that participants believe AI-generated content might be another form of advertising, as commented by P14 *“I feel the same way with Tiktok when you have, like an influencer that is promoting a product. But then they have the note that says, this is sponsored content. I immediately skip it.”* One participant, P14, had negative feelings toward AI-generated content regardless of the labeling: *“Instagram, where I feel like it’s fake and more and more people are seeking authenticity. So, even if it is telling you that it’s AI generated, It feels fake.”*

4.2.4 Content Creator and Platform. One participant mentioned that AI labeling reminds them of the level of agency in the content creation process. P11 commented *“I think it’s about the how much agency the person has to edit everything.”* However, P10 noted that if the content is generated by AI, the content creator has nothing to do with it: *“if you have a tag that, says AI generated, I would assume you had nothing to do with it.”* Regarding platforms, several participants valued their transparency in adding AI labeling. As P12 commented, *“I appreciate that platforms would put labels like this on things because I know, especially for older generations, they don’t entirely understand what is AI and what isn’t, so putting labels on posts like this is helpful.”*

4.2.5 Reaction to the labeling itself. Regarding AI labeling, some participants value the content being transparent as commented by P2 *“I can tell it is AI. So, for me, if they actually had these tags, I would respect the posts for being transparent.”* P1 values labeling because it helps her not to get confused: *“I can take a step back with that tag, so I don’t get immersed in it and think about this. I think the flag makes me not get confused about the things and the reality.”* However, some participants mentioned that the flag confuses users because they do not know what portion of the content is generated or modified by AI. P11 said that *“When I see the label ‘AI generated,’ I’m usually a little bit confused because it could mean that the voice is AI generated, which means it’s reading off some script, and someone’s just using it for convenience.”*

4.3 Why are some labels preferred over others?

4.3.1 Labeling requires clear message. We asked our participants which types of labeling they prefer for AI labeling. Examples included AI-generated, AI-modified, AI-created, AI Info, and Made with AI.

- “AI-modified”, in cases where humans are involved in the content creation

Among the current practices of AI labeling on social media, including “AI-generated,” “Made with AI,” “AI Info,” “AI-modified,” and “AI-created” (Figure 2). Participants indicated that their preferences depend on the context: whether the content is fully generated by AI or whether humans are involved in the creation process. For AI-generated content that also involves human input, participants preferred the label “AI-modified” as it offers greater clarity. P13 commented: “If it’s a real person who changed the background, I would like ‘AI-modified’ the best.” Similarly, P10 noted: “AI-modified is probably my favorite just because it doesn’t feel fully AI, and there is some human aspect to it.”

- “AI-generated” or “AI-created”, in cases where the content is fully created by AI

When the content is fully created by AI, such as virtual influencers, participants favored both “AI-generated” and “AI-created,” as they felt that these terms convey that the content is entirely AI-driven with minimal human involvement. An example quote from P11: “‘AI-created’ is on the extreme end, meaning that everything here is from AI.” Additionally, several participants expressed greater familiarity with “AI-generated,” noting that they had encountered it more frequently than other labeling terms. As P1 commented: “‘AI-generated’ sounds the most natural to me because we’re in an age of generative AI, so I’m very used to hearing that term.” However, after reviewing different AI labels, some participants began questioning the specific implications of “AI-generated.” For example, P3 noted: “‘AI-generated’ seems to need to divide the levels.” Similarly, P4 suggested that the label “should clarify how much of the content is created by AI.” “AI Info” and “Made with AI” were the least favored by our participants, as they found these terms ambiguous. For example, P14 commented: “I would rank ‘AI Info’ at the bottom because, for me, it doesn’t really give much information or make any sense, and the same with just ‘Made with AI.’ I’d rather it be more specific.”

4.3.2 Who should be labeling agency? Reasons why self-labeling may be preferable to platform labeling. Most participants prefer content creators to disclose AI-generated content upfront rather than having it flagged by the platform. The main reason was that participants would perceive creators’ self-disclosure as more genuine and transparent. As P12 commented: “A creator flagging their own content as AI feels more genuine to me ... and very transparent with their audience.” In contrast, when the content is flagged by the platform, it can imply that the creators are not being genuine and may be attempting to hide something from the audience, which participants noted could feel suspicious and even annoying. An example quote from P14 echoed this sentiment: “I feel like I would almost feel like the person who posted it and created the content was trying to hide the fact that it was AI-generated if they didn’t disclose it.” Additionally, participants expressed greater trust in content

flagged by the creator, particularly when it involves realistic images. As P2 noted: “Some AI content can actually look fairly realistic to the untrained eye. When the platform flags it as AI-generated, it almost makes you feel more suspicious. For me, it would make the post more credible if the creator reveals it.” However, creators’ self-disclosure can hold little value without a clear explanation of which parts of the content were generated by AI. As P11 noted: “I would like the content creator to tell me exactly which part AI is being used on. The ambiguity automatically makes me to extreme skepticism, making me think, ‘Oh, well, I guess everything’s AI-generated,’ and then I’ll just scroll past thinking, ‘Ugh! Annoying.’” In addition to the source of labeling, the nature of the content also influences how participants perceive the labels. For lighthearted topics, such as humorous content, the source of the disclosure may be less important. However, for serious content, the issue extends beyond who discloses AI involvement; even when creators self-disclose, it is often not deemed acceptable by audiences. As P13 explained: “like fitness content, whether it was labeled by the platform or by the influencer, I would probably roll my eyes and scroll past it. But if it was humorous, like an animated joke video, I wouldn’t take that as seriously, so I wouldn’t care as much.”

5 Discussion

Users’ perception of AI-generated content differs from their perception of AI-labeled content on social media. When asked what “AI-generated” means to them, participants highlighted positive aspects, such as humans taking the lead with initiative while AI assists in enhancing human capabilities. This is similar to Jakesch et al’s findings, where users did not show more negative feelings toward AI-generated user profiles when it was clearly identified as AI-generated, as opposed to a human-AI mixed condition [8]. However, when asked about AI-labeled content (and being exposed to the example of AI labeled social media post), participants expressed negative feelings with more diverse reasons. First of all, when AI usage in the content is provided as a cue, users tend to interpret it within the social media ecosystem, where both AI and non-AI content coexist. Additionally, due to the unclear usage of AI in the content, users’ perception of the content’s source is affected. Without labeling, the source of content is attributed to the account owner. However, when exposed to AI-labeled content, AI functions as an additional source layer. Contrary to our expectations, users’ perceptions of the content, content owner, and platform regarding AI-labeled content were not significantly different, suggesting that some individuals engage more thoughtfully (e.g., focusing on the content creator) when using social media, while others primarily concentrate on enjoying the content itself. Altay and Gilardi’s study sheds light on this finding because it found that news headlines labeled as AI-generated reduce perceived accuracy and sharing willingness, regardless of content source (human-generated or AI-generated) and content veracity (true or false)[1].

The user’s perception of AI-labeled content also depends on the type of content and account. Some participants mentioned that in some social media accounts, they expect more integrity such as photographer or influencer accounts. However, they mentioned that if it is an account focusing on comedy, they don’t care whether it is generated by AI or not. This is aligned with previous findings

that when content is consumed for entertainment, its enjoyment or creativity may hold more significance than its trustworthiness [19] [10].

Participants showed a stronger preference for content creators to disclose their AI usage directly while expressing greater skepticism and indifference toward content that is AI-labeled by the platform. According to the MAIN model [22], the manner in which a media interface presents sources of information serves as an agency cue that activates various cognitive heuristics (mental shortcuts), such as the machine heuristic (e.g., perceiving content as synthetic rather than authentic) and the bandwagon heuristic (e.g., viewing content as trendy or popular). In the context of AI labeling practices, the attribution of who decides to apply an AI label (content creator vs. the platform) elicits distinct psychological responses. This could be due to individuals' psychological distance from the source, such as news aggregators versus news media organizations that produce news [9]. When consuming news from aggregators like Google, individuals tend to rely on proximal sources (e.g., the news portal) to assess news credibility, rather than distal sources, such as The New York Times, which operate at a lower hierarchical layer in the information delivery process [9]. Following this logic, content creators act as proximal sources of information dissemination on social media, making their self-labeling of AI usage more influential in shaping users' perceptions compared to platform-applied labels. Another reason for the preference in AI labeling sources may relate to normative theory. People value honesty and authenticity in interpersonal communication, including AI-mediated communication [18]. Research shows that secret AI usage is seen as less acceptable than open disclosure [18]. When creators fail to label AI-generated content, which is later flagged by the platforms, it highlights the lack of transparency, reducing trust in both the content and the creators.

6 Implications

Social media designers need to understand that adding AI labeling is not always beneficial. They must consider how this labeling will affect users' perception of content in general, not just for labeled content but also for content without labels. Designers should also recognize that motivating content posters to label their content might be a more effective way to ensure positive effect of AI labeling. Therefore, we argue that designers should focus on how to encourage posters to add the flag when uploading their content. Just as Amazon tries to filter out fake reviews through its internal system, social media platforms could consider detecting AI usage when individuals attempt to post content. They could then provide clear warnings that, without self-flagging, the content will be flagged, and inform posters of the potential detrimental effects of platform flagging. Furthermore, we have learned that the labeling message should be clear and more specific. We suggest that labeling be divided into several categories based on how much of the content is generated by AI. Finally, we found that users' perception of AI labeling depends on the type of content and accounts involved. Designers should consider differentiating labeling policies based on content and account type.

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A Appendices

A.1 Research Protocol

- First, I would like to know what “AI-generated” really means to you. Could you please explain how you interpret “AI-generated” in the content?

- AI labeling is a technological feature on the content in social media like TikTok, Instagram and Facebook. Please see the examples below.
- How does this AI labeling affect your perception toward the content, content-creator (account holder) and the platform? Do you think your perception would be different without this labeling?
- What kind of labeling do you prefer? Please rank order them and explain why. 1) AI-generated, 2) Made with AI, 3) AI Info, 4) AI modified, and 5) AI-created.
- There are two types of AI labeling, depending on who adds the label to the content. One involves the account holder disclosing that the content is generated by AI, while the other involves the platform identifying AI usage in the content and adding the label. If you are exposed to two different types of labeling—disclosed by the creator vs. detected by the platform—in the same content, how would you perceive the content differently?

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